PSF & Convolution

Lec 7-1

M = magnification.

Finite aperture of objective causes ~~diffraction~~ diffraction.

If M = 100 (i.e., 100×)

and camera pixel



in specimen plane.

$\vec{r} = (x, y)$ = image coordinate (in specimen plane)

$\vec{r}_0$ = molecule's true lateral position ~~in~~ xy plane

Normalized intensity PSF:

Def^n: $h(\vec{r} - \vec{r}_0)\, d^2r = P(\text{a detected photon lands in } d^2r \text{ near } \vec{r} \mid \vec{r}_0)$

$$\int h(\vec{r})\, d^2r = 1$$

over all pixels

$h(\vec{r})$ = Probability density → $\frac{1}{\text{area}}$

one detected event = creation of one photoelectron.

Fraction of photons captured by a region A = $\int_A h(\vec{r} - \vec{r}_0)\, d^2r$.

$q(t) \equiv$ expected detected rate across the full image plane

↳ $\frac{\#\text{ of photoelectrons}}{\text{time}}$

∴ $q(t)\,dt$ = # of detected events within dt.

$h(\vec{r} - \vec{r}_0)\, d^2r$ = fraction expected near $\vec{r}$



$\therefore$ $q(t)\,dt$ = mean total counts (# photons detected) within dt.

$q(t)\,dt \cdot h(\vec{r}-\vec{r}_0)\,d^2r \equiv \lambda_s(\vec{r},t)\,d^2r\,dt \Rightarrow$ $\boxed{\lambda_s(\vec{r},t) \equiv q(t)\,h(\vec{r}-\vec{r}_0)}$

- $h(\vec{r}-\vec{r}_0)\,d^2r$: fraction of events within d^2r.
- $\lambda_s(\vec{r},t)$: expected (= mean) detected signal rate per unit area (~ flux of events).

1) Assume shift-invariance:

moving the molecule moves its image by the same displacement without changing the PSF's shape

2) Also assume: independent fluorophores contribute linearly.

$\therefore$ $\vec{r}_j(t)$ = molecule j's position

$q_j(t)$ = expected detection rate for molecule j.

$\Rightarrow$ $\boxed{\lambda_s(\vec{r},t) = \sum_j q_j(t)\,h(\vec{r}-\vec{r}_j(t))}$

- $\lambda_s(\vec{r},t)$: rate/area
- $\vec{r}_j(t)$: molecule-j could move (so t-dependance)

$\Rightarrow$ PSFs of two sufficiently nearby fluorophores will overlap

$\Rightarrow$ Cannot resolve their separation distance or even that there are two (or more) fluorophores there!

Replace individual molecules by a source distribution : $f(\vec{r}', t)$

↑ prime marks source coordinate.

$f(\vec{r}', t)$ = expected (= mean) detected rate per unit specimen area before microscope spreads photons.

$\Delta A'$ ≡ small patch area.

∴ $f(\vec{r}_j', t)\,\Delta A'$ = rate supplied by that patch.

ΔA = small target area (target patch) in image plane (on camera pixel) near image coordinate $\vec{r}$.

∴ $h(\vec{r} - \vec{r}_j')\,\Delta A$ = fraction of source patch's signal that reaches the target patch

∴ $f(\vec{r}_j', t)\,\Delta A'\,h(\vec{r} - \vec{r}_j')\,\Delta A$ = Contribution by patch $\Delta A'$ to count rate at ΔA pixel.

Let $R_{\Delta A}(\vec{r}, t)$ = total expected count rate in the target patch.

then: ① Sum the source contributions:

$$R_{\Delta A}(\vec{r}, t) = \sum_j f(\vec{r}_j', t)\,\Delta A'\,h(\vec{r} - \vec{r}_j')\,\Delta A$$

② Next, divide by target area to get rate density

$$\lambda_s(\vec{r}, t) = \frac{R_{\Delta A}(\vec{r}, t)}{\Delta A} = \sum_j h(\vec{r} - \vec{r}_j')\,f(\vec{r}_j', t)\,\Delta A'$$

③ finally, make ΔA & $\Delta A'$ infinitesimal:

$$\lambda_s(\vec{r}, t) = \int h(\vec{r} - \vec{r}')\,f(\vec{r}', t)\,d^2r' \equiv (h * f)(\vec{r}, t)$$

rate/area ↑ ↑ convolution.

Convolution here means: for each source position, place a copy of the PSF there (i.e., $h(\vec{r}-\vec{r}')$), weight it by the source brightness (i.e., f), and add all the copies.

For point emitters: $$f(\vec{r}', t) \equiv \sum_j q_j(t)\, \delta^{(2)}(\vec{r}' - \vec{r}_j(t))$$

↳ from this, we recover the $\sum_j$ from the $\int$.

Caution: for a thick specimen, sources at different depths have different PSFs. In that case, we must integrate over the depth z' as well.

i.e., $$\lambda_s(\vec{r}, t) = \int h(\vec{r}-\vec{r}'; z')\, f_3(\vec{r}', z', t)\, d^2r'\, dz'$$

λ_s = $\frac{\text{rate}}{\text{area}}$ for every position & time.

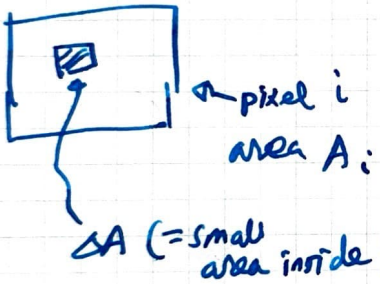
But camera records something simpler: one # for each pixel during an exposure.

Let $A_i \equiv$ specimen-plane area corresponding to pixel i.

Let frame n start at t_n and last for time T.

Let $\lambda_b(\vec{r}, t) \equiv$ expected detected optical background rate per unit area.

Background photons may come from out-of-focus fluorophores or other fluorescent material in the specimen.



$\Delta t \equiv$ short time within one exposure.

~~then expected count from that pixel w. within Δt is~~

~~mean count in ... Δt~~

Then, expected count (mean) in pixel i =

$$\sum_{k} \sum_{\text{area patches}} \left[\lambda_s(\vec{r}_\ell, t_k) + \lambda_b(\vec{r}_\ell, t_k)\right] \times \Delta A \, \Delta t$$

k: time bins

~~Shrink time bins~~ and

shrink time bins and area patches to get:

$$\mu_{i,n} = \int_{t_n}^{t_n+T} dt \int_{A_i} d^2r \left[\lambda_s(\vec{r},t) + \lambda_b(\vec{r},t)\right] + d_i T$$

$\mu_{i,n}$: Counts

i: pixel i

n: time frame n (i.e., n^{th} exposure)

T: exposure time.

d_i: dark current in detector (camera)

$d_i \equiv$ # e^-s / time

↳ generated w/o photons.

How to use $\mu_{i,n}$:

For one stationary molecule (fluorophore) with constant detected rate q,

define: $p_i(\vec{r}_0) = \int_{A_i} h(\vec{r}-\vec{r}_0)\, d^2r$, $N \equiv qT$

p_i: probability that detected signal photon lands in pixel i.

N: total # of photons expected in time T (when summed over all pixels)

then: $\mu_i = N p_i + B_i + d_i T$

μ_i: mean # of detections in pixel i during one exposure

B_i: optical background count expected during this exposure

Concretely, we use as PSF : Gaussian

$$h(x,y) = \frac{1}{2\pi s^2} \exp\left(-\frac{x^2+y^2}{2s^2}\right)$$

with fluorophore at center of pixel:



$$p_0 = \left[\int_{-a/2}^{a/2} \frac{\exp\left(\frac{-x^2}{2s^2}\right)}{\sqrt{2\pi}\, s}\, dx\right]^2$$

In this case, we get these concrete #s :

say $s = 150\,nm$
$a = 100\,nm$
then $p_0 \approx 0.06818$
so if $N = 1000$ signal counts,
we expect : $\mu = 1000 \times (0.06818)$
≈ 68 counts in this particular pixel.

Qu: We have described how a molecule produces a blurry spot. Now suppose we know this spot comes from one molecule. How precisely can we determine its center?

Key Idea: Width of the spot and the uncertainty in its center are different quantities.

Key Idea: More photons help

Each photon lands at a random position x_i, with mean x_0 and variance s^2.

Then $\hat{x}_0 = \frac{1}{N}\sum_{i=1}^{N} x_i$ $\quad \mathrm{var}(\hat{x}_0) = \frac{1}{N^2}\sum_i s^2 = \frac{s^2}{N}$

$$\therefore \boxed{\sigma_{loc} = \frac{s}{\sqrt{N}}}$$

$\Rightarrow$ A spot with $s \approx 100$ nm can give about 1-nm ideal precision with 10,000 detected photons.

Assumptions here: . one isolated emitter
. Independent photons
. negligible motion during the exposure.

Caveats: . real cameras fit the PSF to pixel counts
. the above ideal calculation explains the scaling.
. Background & systematic errors worsen the result.